

L'ASSOMPTION, QC, Canada

WMO: 715240

Lat: 45.8094N

Lon: 73.4347W

Elev: 21

StdP: 101.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-22.6	-30.5	0.2	-24.9	-27.8	0.3	-22.0	10.4	-2.1	9.1	-3.0	1.9	240	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.5	22.0	28.8	20.9	27.3	20.2	23.5	28.2	22.5	26.7	21.5	25.5	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	16.7	26.0	21.0	15.7	24.9	20.1	14.8	24.0	70.3	28.1	66.2	26.8	62.8	25.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
8.3	7.2	6.3	-29.9	33.0	3.4	1.5	-32.3	34.1	-34.3	35.0	-36.2	35.8	-38.7	36.9		
			-30.1	25.5	3.4	1.2	-32.6	26.4	-34.6	27.1	-36.5	27.8	-39.0	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-10.3	-8.7	-2.8	5.6	13.4	18.4	20.8	19.8	15.5	8.4	1.5	-5.8
	DBStd	12.00	7.30	6.31	5.99	4.52	4.28	3.70	2.92	3.03	4.16	4.36	4.92	6.44
	HDD10.0	2543	629	523	397	144	14	0	0	0	4	85	257	489
	HDD18.3	4582	888	756	654	382	162	43	9	20	104	309	505	748
	CDD10.0	1230	0	0	1	12	120	253	335	304	169	34	2	0
	CDD18.3	228	0	0	0	1	9	46	86	65	20	1	0	0
	CDH23.3	1857	0	0	1	8	122	408	688	478	149	4	0	0
	CDH26.7	465	0	0	0	1	29	113	182	107	33	0	0	0
Wind	WSAvg	2.8	3.0	3.0	3.1	3.3	3.0	2.7	2.5	2.3	2.4	2.7	2.9	2.9
Precipitation	PrecAvg	1017	74	55	63	83	90	100	97	103	88	96	79	89
	PrecMax	1227	142	118	131	155	187	223	166	182	173	173	141	168
	PrecMin	785	22	25	18	11	47	29	61	49	28	28	12	19
	PrecStd	118	31	22	29	38	34	45	26	31	36	38	29	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	5.7	14.1	23.8	30.0	32.0	32.8	31.3	30.4	23.4	16.3	9.0
		MCWB	6.7	3.5	8.7	15.1	19.9	22.4	23.8	22.4	21.9	17.1	13.7	7.3
	2%	DB	4.0	3.8	9.3	19.4	26.6	29.7	30.4	29.3	27.0	20.2	13.1	6.0
		MCWB	2.7	2.2	5.4	11.5	17.1	20.9	22.4	21.5	20.5	15.2	10.7	4.8
	5%	DB	2.2	2.3	6.6	16.1	24.0	27.7	28.6	27.7	24.7	17.4	10.6	3.8
		MCWB	1.3	1.1	3.6	9.8	15.6	19.8	21.4	20.7	19.2	13.4	8.4	2.7
	10%	DB	0.3	0.9	4.7	13.2	21.4	25.6	27.1	26.0	22.7	15.2	8.4	2.1
		MCWB	-0.5	-0.1	2.3	8.0	14.6	18.5	20.5	19.9	18.0	12.0	6.4	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	4.2	10.2	16.4	21.4	24.1	25.3	24.4	23.1	18.4	14.3	7.8
		MCDB	7.3	5.0	12.9	21.9	27.5	29.8	31.0	28.9	27.8	21.9	15.4	8.7
	2%	WB	3.0	2.4	5.9	12.8	19.1	22.4	23.9	22.9	21.5	15.7	11.2	4.9
		MCDB	3.5	3.4	8.3	16.9	23.8	27.6	28.4	27.3	25.1	19.0	13.0	5.8
	5%	WB	1.3	1.2	4.0	10.3	17.3	21.1	22.6	21.9	20.2	13.9	8.5	2.8
		MCDB	2.1	2.2	6.0	15.0	21.8	25.4	26.6	25.8	23.8	16.5	10.3	3.7
	10%	WB	-0.4	0.0	2.6	8.5	15.5	19.9	21.6	21.0	18.8	12.3	6.6	1.2
		MCDB	0.4	0.9	4.4	12.5	19.7	24.0	25.6	24.6	22.0	14.5	8.2	2.0
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	9.2	10.0	11.0	10.7	10.6	10.8	10.9	9.0	7.3	7.6
		MCDBR	9.2	8.6	10.4	15.9	15.1	13.5	12.2	12.3	12.7	12.3	10.5	7.6
		MCWBR	8.9	7.7	7.5	9.3	7.7	6.5	6.0	6.0	7.1	7.9	8.6	6.9
	5% WB	MCDBR	9.0	8.2	9.5	13.4	12.5	11.3	10.6	10.5	11.2	10.8	9.9	7.3
		MCWBR	8.7	7.7	7.3	8.9	7.7	6.4	5.9	5.8	6.8	7.7	8.7	7.0
Clear-Sky Solar Irradiance	taub	0.255	0.289	0.335	0.364	0.392	0.437	0.428	0.415	0.376	0.337	0.307	0.293	
	taud	2.347	2.238	2.241	2.341	2.328	2.209	2.240	2.275	2.372	2.486	2.481	2.290	
	Ebn at Noon	864	886	888	894	876	834	837	833	838	828	793	754	
	Edh at Noon	73	101	118	117	123	140	134	124	103	80	67	71	
All-Sky Solar Radiation	RadAvg	1.44	2.37	3.62	4.40	5.12	5.62	5.62	4.95	3.91	2.26	1.36	0.99	
	RadStd	0.17	0.25	0.31	0.49	0.50	0.42	0.44	0.23	0.32	0.19	0.17	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page